http://193.6.135.152/exam.progcont.eu/

Root

Stop the countdown in the GRUB menu by using arrows and choose the first one.

type "*rd.break*" before "root=" in line 4 and ctrl + x.

mount -o remount, rw /sysroot "changes the ro i.e. read-only permission to rw i.e. read-write permission.

chroot /sysroot/ "opens a new shell in which the contents of sysroot act as top priority"

passwd "gives you the ability to change the password of the current user i.e. the system administrator"

Due to the complex nature of the "passwd" cmd, we need to restore the missing attributes used for the password.

touch /.autorelabel "fixes missing attributes"

"*exit*" cmd to exit the shell and then again to exit the emergency shell.

Port forwarding host port 10022

SSH putty hostname:local host port:10022

ls 'show files in current folder'

ls -a 'show hidden files'
    -l 'see more details'

-i 'shows unique inode no.'

-lai 'in any order combines all'
-d 'shows the properties of the directory'
pwd 'print working directory'
echo 'printf'
echo hello > message.txt 'printf into a file'
stat 'gives information abt the file'
d 'appears for directories in the beginning'
- 'appears for files in the beginning'
rwx rwx rwx 'permissions'
First 3 is owner user denoted by 'u'
Second 3 is owner group denoted by 'g'
Third 3 is other denoted by 'o'
This system is called 'DAC' model.
chmod (new access settings) file....
Group of user - ugo
Actions- set grant revoke (=,+,-)
Roles (r w x)
Example : chmod o+x message.txt
chmod ug-r message.txt
chmod u=rw, go=r message.txt

To check the permissions : (example)

ls -ali /home/student/message.txt

000 - - - (0)
001 - - x (1)
010 - w - (2)
011 - w x (3)
100 r - - (4)
101 r - x (5)
110 r w - (6)
111 r w x (7)

Example usage: chmod 000 message.txt
(No permissions to read write or execute)

Putty

First in the virtual machine's *setting* menu go to the *network* tab then go to *advanced* and click on *port forwarding* add a new instance and change the *guest port to 22* and *host port to 10022.* Now start the machine, its enough to start in headless mode or normal mode

doesn't matter, now go to putty and put
host name as localhost

and *port as 10022*, and make sure the *SSH conection type* is chosen and click open to open the terminal on putty and login as root.

EXAM

After logging in as root type in *exam* to see the 2 different usages of it, one is to prepare the system for the exam and the next one is to submit your answers to the exam site. The exam site will be given on elearning from where u can login using ur edu id into the system and start the exam, there is just about enough time to finish the exam. A comand will be available for you after you start the exam on the site copy it and paste it in ur terminal and click enter to prepare your virtual machine according to your exam, its unique for every student. The two commands are the same as we discussed above previously, its just the unique id that changes.

One to initiate and the other to submit. You can submit after each answer for partial marks.

Questions

1. Find the directory in /root/86344/, where the owner is not the root user. Write the (absolute) path name to the file: /root/86344.ans

find /root/86344/ -not -user root
>/root/86344.ans

2. Create a new user with name user86315 and lock the password of the user!

useradd user86315
passwd -l user86315

3. Who signed the file /root/contract-86302.txt, Anne, Bob, Carol, Daniel or Eve?

The signature was created with the help of the SHA 256 hash algorithm and stored in the file: /root/contract-86302.sha256.signature

The public key of Anne is in /root/Anne-86302.pub.pem

the public key of Bob is in /root/Bob-86302.pub.pem

the public key of Carol is in /root/Carol-86302.pub.pem

the public key of Daniel is in /root/Daniel-86302.pub.pem

the public key of Eve is in /root/Eve-86302.pub.pem

Write the answer (the name) to the file: /root/86302.ans

openssl dgst -verify /root/

Anne-86302.pub.pem -signature /root/contract-86302.sha256.signature -sha256 /root/contract-86302.txt

Check similarly using public keys of every one, basically change the name after -verify and cycle through every name i.e. Anne, Bob, Carol, Daneil, Eve unitl you get Verified Ok *instead of* Verification Failure*.*

echo Carol > /root/86302.ans

4. Download the file 5d3a8e1b with the help of the command below.

curl -o/5d3a8e1b https://progcont.hu/docs/IT-Security/loop/5d3a8e1b

The file contains an encrypted filesystem in LUKS format. The password is: 5d3a8e1b. Copy the content of the filesystem to the directory: /root

Copy the cmd and paste it into as step one as instructed, next

losetup -f /5d3a8e1b
cryptsetup luksOpen /dev/loop0 loop
Now enter the password in the prompt
5d3a8e1b
mount /dev/mapper/loop /mnt
ls mnt
You will get an output such as 070f lost+found, now to copy:
cp /mnt/070f /root
Or
cp -r /mnt/lost+found/ /root
Or
cp /mnt/* /root

5. Decrypt the /root/86261.enc file by using 86261 as the key and 0 as the initialization vector. The file was encrypted by using the aes-128-cbc algorithm. Save the result as /root/86261.txt.

openssl aes-128-cbc -d -in /root/86261.enc -K 86261 -iv 0 > /root/86261.txt

6. Decrypt the /root/86257.enc file by using the 86257 password and the des-ede algorithm. Save the result as/root/86257.txt.

openssl des-ede -d -in /root/86257.enc -k 86257 > /root/86257.txt

7. Sign the /root/86247.txt file with the following private key. Use the SHA 256 hash function. Save the signature as/root/86247.sign!

--BEGIN PRIVATE KEY-----

MIICdgIBADANBgkqhkiG9w0BAQEFAA
SCAmAwggJcA
6JBmdQL4fJmDZolEKJIojg9IapEI95SV5
gsZb9+K6.
xDaWM0rN/wvM+MzMDoz2I0ep6I/VQM
Fvq9/kL0iwhl
iAtr9xl+3Gx6VIZ5ZgzpSH0eFspTAgMBA

AECgYBpf/gYrIJ+8RvQOALRQeHz2iNk
5G/oW8JaEs1lYHaD10j mi

JbcZaX0N6pY11BZOVaGgSMWR2/frQV
VPH9pdVk5l
yeMUer8bh5/GSJL8YQJBAONwyYVu0
Qo2dkVZTtGorr
YvzPdABDhjY+kjeA0LhT1xVwFr7Mgek
Gks/EgbKlbl
gXJTNNqbNgLf3+tQBNru8HCo1Ed4HPs
cx5tpfZHie)
mKVDAkBiEuyTI0hvwvRVCyG1vqsWW
NOXBDUILAHN7)
qJ5YHa8Ni3wDYmRSe4crGG1k/3jBAkE
Ass3/mGZril
PoB4sjCmp2jQVC0ko8AXwqG0MKhg85
KfyJd7VqlLW
Dmqu2Q+KrIyYSDfWwdFDda6qkH9QH
oTQ2aE6RtkShl 200qC6Dvgv2y/A==

-----END PRIVATE KEY-----
umask 077

cat >k

Paste the block of key starting from begin private key to end private key "Enter and then ctrl+d" openssl dgst -sign k -sha256 -out /root/

86247.sign /root/86247.txt

8. Set the permissions of the /root/86237.txt file to read-only.

chmod 444 /root/86237.txt

9. Give read-only permissions to the operator user on the /root/86224.txt file with a new USER ACL entry. (Do not change anything else about the file and do not give or take any other permissions.)

setfacl -m u:operator:r /root/86224.txt

10. Set owner of the /root/86211.txt file to the operator user. (Do not change anything else about the file.)

chown operator /root/86211.txt

11. Create a file with name /root/86198.dat and with a size of 861980 bytes.

```
dd if=/dev/zero of=/root/86198.dat
bs=861980 count=1
```

12. Generate an 1024 bit RSA key and save it as /root/86191.pem as a PKCS8 private key.

openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:1024 -out /root/86191.pem

13. Create a new /dev/sdc1 partition on the drive /dev/sdc. Format the partition with ext2 filesystem and mount it to the /mnt folder in read only mode.

fdisk /dev/sdc

n

"Enter ×5"

w

mkfs.ext2 /dev/sdc1

mount -o ro /dev/sdc1 /mnt

mount

14. Allow the owner of the following private key to log in as the root user.

----BEGIN PRIVATE KEY--

MIICdwIBADANBgkqhkiG9w0BAQEFAA
SCAmEwggJdA
GPNsk+p9HFUTwSM1ka80tX0Hu80+bq
Ef86g+gMOB!
v4ijljj/6xJcCqPP2LVeYVM1aDWjYADHm
HRFOJwLC
u7RSsX9W8mCAqx1xlpfkpiQiiTUSAgM
BAAECgYBJE
EOVlh/BTL+UY65NXFg7jneCo2zdnPKV
JwtlodtpZ+(
D70FVscj9TSIzP1lHBNcWdt1ckZPR1A
Dnauueg818
H0cY6qRQjkNgipJVCQJBA02RAVk21E
34jE6+6w2/jl
2DDkBxVHKYLdtnnL9eX3069T4UKYEZ
W58GkGXgA-
*kfTW7x4pFwgjLEyd+Kzs5LpmU8MxEL
&naXZVHnmy1r

z4qjAkAF1R6MWy1BMM35DPTmZIKG9JA

/+1kyPKsZcl
+ary12w0t0d95A/iibqy781RwzoDAREA
AS4P80w51
b7qMv9aETRajXJ7AXVdnff1r9uyVFydl
W3yMdMxm
tTaU838GJB5L8Afq/HIW+Biw/GmgcuK
MbOPREOdIkl aEUNnE4+0e7tQXE=

-END PRIVATE KEY----

umask

cat >kk

Paste the complete block inclusing the begin private key and end key private key in and then press enter then ctrl+d just like before.

mkdir .ssh

ssh-keygen -y -f kk
>.ssh/authorized_keys

15. Create the user160184 user and generate a new default ssh RSA key (/home/user160184/.ssh/id_rsa) for it

without passphrase. Allows user160184 to log as root with the help of the RSA private

key.
useradd user160184
su user160184
ssh-keygen
"Enter ×3"
ssh-copy-id root@localhost
yes
exam
exit

16. Find the directory in /root/160171/, in which the sticky bit is set. Write the full (absolute) path name to the file: /root/160171.ans
find /root/160161/ -perm /1000
>/root/160171.ans

17. Configure the user account operator so that its password expires on 2021-07-24 and its maximum number of days between password change is 100.
chage operator -M 100

chage operator -l
chage operator -d 2021-04-15

18. Set the permissions of the file /root/160101.txt so that its permissions shows as -r--rws--T in the directory listing.
chmod 3470 /root/160101.txt

19. Delete all ACL entries of the file /root/160093.txt
setfacl -b /root/160093.txt

20. Set ownership of the file /root/160076.txt so that the owner will be operator and the owner group will be users. (Do not change anything else about the file.)
chown operator:users /root/160076.txt

21. Copy the first 70 byte of the file /root/160067.dat to the file /root/160067.part.

```
dd if=/root/160067.dat of=/root/160067.part bs=70 count=1
```

22. Generate an 1024 bit RSA key and save it as /root/160060.pem as a PKCS8 private key.
openssl genpkey -algorithm RSA -pkeyopt rsa_keygen_bits:1024 -out /root/160060.pem

23. Create a new crypto_LUKS encrypted xfs filesystem on /dev/sdc and mount it to the /mnt directory.
cryptsetup luksFormat /dev/sdc
YES
appleAPPLEapple
appleAPPLEapple
cryptsetup luksOpen /dev/sdc x
appleAPPLEapple
mkfs.xfs /dev/mapper/x
umount /mnt/
mount /dev/mapper/x /mnt

24. Who signed the file/root/

contract-160147.txt, Anne, Bob, Carol, Daniel or Eve?

The signature was created with the help of the SHA 256 hash algorithm and stored in the file: /root/contract-160147.sha256.signature

The public key of Anne is in /root/Anne-160147.pub.pem

the public key of Bob is in /root/Bob-160147.pub.pem

the public key of Carol is in/root/Carol-160147.pub.pem

the public key of Daniel is in/root/Daniel-160147.pub.pem

the public key of Eve is in /root/Eve-160147.pub.pem

Write the answer (the name) to the file: /root/160147.ans

openssl dgst -verify /root/Anne-160147.pub.pem -signature /root/contract-160147.sha256.signature -sha256 /root/contract-160147.txt
Check similarly using public keys of every one, basically change the name after -verify and cycle through every name i.e. Anne, Bob, Carol, Daneil, Eve unitl you get Verified Ok *instead of* Verification Failure*.*
echo Eve > /root/160147.ans